Medical Report Management & Distribution System on Blockchain

1) Background/ Problem Statement

In the current era of advanced technology, patient data such as name, personal information, and ailment condition are highly insecure and frequently violated. Patients' Electronic Health Records (EHR) are easily accessible to unauthorized external parties and are easily accessed by hackers. This results in an imbalance between data accessibility and security. This is something that blockchain technology can help with.

Our project, Medical Report Management & Distribution System on Blockchain, is designed to secure medical reports and provide a safe way of accessing the clinical data of patients for their caretakers and doctors using cryptography and decentralization.

The proposed system also maintains the balance between data accessibility and security. The proposed framework helps doctors, patients and caretakers to securely store and access patients' medical data in EHR.

2) Working of the Project

This Python-based project is designed to maintain and distribute patients' medical records efficiently and effectively. It consists of 3 modules: Hospital, Pharmacy and Pathology Lab.

Hospital administrators can manage all the data related to Pharmacies, Pathology labs and Patients. Pharmacy can add and view patients' bills. The Pathology Lab add and view patients' reports in this system.

In this project, Html, CSS and JavaScript are used in the front end and Dotnet is used in the back end. The Database used is MSQL and the IDE used is Visual Studio.

3) Advantages

- It is easy to maintain.
- The system is user-friendly.
- Doctors can keep track of their patient's medical.
- The system facilitates accurate decision-making in diagnosis by accessing up-to-date medical history.
- Patients' medical records are maintained and distributed securely.

4) System Description

The system comprises 3 major modules with their sub-modules as follows:

❖ **Hospital:**

- Login:

  The hospital administrator can log in to the system using a username and password.

- Manage Pharmacy:

  They can add, update, delete and view Pharmacies in the system.

- Manage Pathology Lab:

  They can add, update, delete and view Pathology Labs in the system.

- Manage Patient:

  They can add, update, delete and view patients in the system.

  They can view the Pharmacy bills of the patients.

  They can view Pathology Lab reports of the patients.

They can check out the status of any fabrication in a bill or lab report.

❖ **Pharmacy:**

- Login:

  The Pharmacy administrator can log in to the system using a username and password.

- Add Bill:

  They can choose each patient and add the bills accordingly.

  They can view bills by using a filter - by patient or date.

❖ **Pathology lab:**

- Login:

  The Pathology Lab administrator can log in to the system using a username and password.

- Add Report:

  They can choose patients and add their reports with report type.

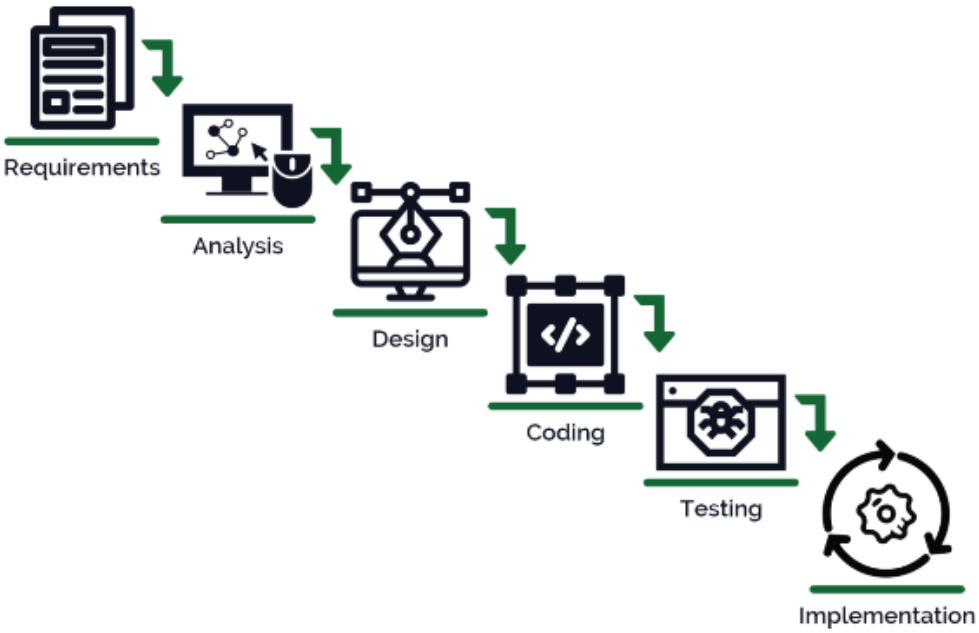
  They can also add values to the report.

- View Reports:

  They can view patients' reports using a filter - by patient or report type.

5) Project Life Cycle

The waterfall model is a classical model used in the system development life cycle to create a system with a linear and sequential approach. It is termed a waterfall because the model develops systematically from one phase to another in a downward fashion. The waterfall approach does not define the process to go back to the previous phase to handle changes in requirements. The waterfall approach is the earliest approach that was used for software development.



6) ***System Requirements***

I. ***Hardware Requirement***

i. **Laptop or PC**

- Windows 7 or higher
- I3 processor system or higher
- 4 GB RAM or higher
- 100 GB ROM or higher

I. Software Requirement

ii.Laptop or PC

- Visual Studio 2019
- SQL Server Management Studio latest

7) Limitations/Disadvantages

The administrators have to enter correct details otherwise it can lead to misleading outcomes.

8) Application

The project is designed to replace traditional paper-based medical records and to enhance hospital data management in healthcare sectors.

*9) **Reference***

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